Question			Answer	Marks	AO	Guidance
11	(a)		Gradient $n = \dfrac{0.376 - 0.254}{0.146 - (-0.097)} = 0.50$	**M1** **A1**	**1.1a** **1.1**	Allow www. Do not allow for reciprocal of gradient Allow for 0.5 or 0.502…seen
			So $k = 10^{0.303} = 2.0$ (2sf)	**M1** **A1**	**1.1a** **1.1b**	Finding the intercept and attempting to find k Maybe implied by 2 or 2.0 etc Must be 2 significant figures. Allow embedded in an equation if not given explicitly
			Alternative method			
			$0.254 = \log k - 0.097n$ $0.376 = \log k + 0.146n$	**M1**		Setting up a pair of simultaneous equations and attempting to solve. Do not allow for $\log T$ and $\log l$ interchanged
			$n = 0.50$	**A1**		Allow for 0.5 or 0.502…seen
			So $k = 10^{0.303} = 2.0$ (2sf)	**M1** **A1**	**1.1a** **1.1b**	Finding the intercept and attempting to find k Maybe implied by 2 or 2.0 etc Must be 2 significant figures. Allow embedded in an equation if not given explicitly
				[4]		
11	**(b)**		$\log_{10} T = \log_{10} k + \log_{10} l^n$			
			$= \log_{10} kl^n$	**M1**	**1.1a**	Uses laws of logs for powers and products
			So $T = kl^n = 2.0l^{0.50}$	**A1**	**1.1**	FT their n and positive value for k
				[2]		